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CARPOOLING: WHO, HOW AND WHY

ROGER F. TEAL
Department of Civil Engineering and Institute of Transportation Studies, University of California, Irvine, Irvine, CA 92717*

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Abstract—Carpooling is the commuting mode of 18 to 20% of American workers, but relatively little definitive information has been available on who carpools, how and why. Based on data from the 1977–78 Nationwide Personal Transportation Survey, this paper analyzes the characteristics of carpoolers, distinguishes among different types of carpoolers, identifies the key differences between carpoolers and drive alone and transit commuters, describes how commuters carpool, and offers explanations of why commuters carpool. The paper also addresses the issue of the feasibility of a substantial increase in carpool mode share.

INTRODUCTION

Carpooling is the second most popular mode of commuting to work, and probably the least understood. Despite a decade long emphasis on increasing the amount of carpooling and other forms of commuter ridesharing, surprisingly little comprehensive and definitive information exists on carpooling, as well as on the realistic potential for significant increases in the use of this mode. Based on an analysis of the 1977–78 Nationwide Personal Transportation Survey (NPTS), this paper represents a step in the direction of understanding carpooling more completely by identifying the types of individuals *who* are most likely to carpool, explaining *how* carpooling takes place and addressing the issue of *why* commuters carpool.

The importance of obtaining a better understanding of carpooling should be clear. According to national surveys, 18 to 20% of all workers who commute in a vehicle are carpoolers, or about three times as many workers as use public transit. In addition, most large urban areas have some type of organized ridesharing program which aims at increasing the portion of commuters who carpool. If such attempts are to succeed, an understanding of the carpool market is essential. Moreover, transportation planners require better information on carpooling in order to determine if usage of this mode can be influenced to any significant degree.

Carpooling occupies a rather curious status as a commuting mode, for in some ways it is inferior to both driving alone and public transit riding, whereas in other respects it is superior to both. Carpooling is inferior to driving alone because it requires an increase in travel time due to the need to pick up and deliver carpool members. Probably more importantly, carpoolers suffer from a significant reduction in convenience due to the schedule rigidity which this mode usually entails, as well as a potentially problematic social experience due to the loss of privacy. Compared to public transit, carpooling can also be less convenient *if* high quality transit service is available, as a much greater degree of schedule flexibility is possible on transit when this mode has frequent departures. Transit may also be cheaper than carpooling, particularly for trips into downtown areas of large cities where parking is relatively expensive. In addition, for many people the anonymity of transit riding is a more comfortable social climate than carpooling.

On the other hand, carpooling has important advantages compared to the drive alone and transit modes. Due to cost or vehicle sharing by carpool members, their out of pocket commuting expenses are at least 50% less than driving alone. Although usually less temporally flexible than transit, carpooling is vastly more spatially flexible. It is a potential commuting option whenever two workers live and work in relative proximity to one another, whereas large portions of even densely populated urban areas are effectively inaccessible by transit for the average commuter. In addition, carpooling is a much less time consuming mode than transit. The average transit commuter trip requires about 70% more time than the average carpool trip, even though the latter is somewhat longer in distance (Bureau of the Census, 1979).

The relative importance of these factors can be inferred from actual patterns of modal use. The fact that three times as many persons carpool as use transit indicates that, at the national scale, its superior availability, spatial flexibility and speed more than offset its temporal rigidity disadvantage. Conversely, the approximately four to one ratio between the shares of the drive alone and carpool modes indicates that the convenience advantages of driving alone are a much more powerful influence on the average commuter than are the cost savings of carpooling.

The purpose of this paper is to provide insight into these commuting patterns by analyzing the personal

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characteristics and transportation situations of those workers who carpool. The paper proceeds as follows. First, the results of previous studies of carpooling are reviewed and used to determine which avenues of inquiry are most important for this analysis. Second, the characteristics of carpoolers are described and contrasted of those of other commuters. Third, the subject of how commuters carpool is explored. Fourth, explanations are advanced as to why commuters carpool, focusing on such factors as commuting cost burden, vehicle availability, household vehicle sharing, number of workers in the household, and transit availability and quality. Fifth, discriminant analysis is employed in an attempt to statistically distinguish carpoolers from users of other commuting modes. Finally, the results of these analyses are extrapolated to indicate the realistic expectations for attempts to increase ridesharing.

PREVIOUS STUDIES OF CARPOOLING

Study Findings

Several important studies of carpooling have been conducted during the past decade. Relative to the issue of *who* carpools, most studies conclude that socio-demographic variables are *not* correlated with carpooling propensity. As Hartgen (1977) noted in a review of the carpooling literature, "all researchers agree that whatever demographic relationships exist are extremely weak ones." In a later literature review, Kostyniuk (1982) reached a similar conclusion. The only socio-demographic variables which have been linked to carpooling propensity are lower income (Kendall, 1975; Voorhees, 1973), lower automobile ownership levels (Voorhees, 1973, Kendall, 1975) and multiple worker households (Hartgen, 1977; Brunso *et al.*, 1979), and none of these factors have emerged as consistently important in the majority of studies. For example, a well-designed study of Chicago commuters found that sex, income, education and the number of automobiles and licensed drivers in the household *did not* distinguish between carpoolers and those who drove alone (Horowitz and Sheth, 1978).

On the other hand, there is general agreement that tripmaking characteristics do influence the propensity to carpool. Almost all studies find that carpoolers travel significantly farther to work than do commuters who drive alone (Kendall, 1975; Margolin and Misch, 1978; Brunso *et al.*, 1979; Hartgen, 1977; Richardson and Young, 1982). In addition, a study of Albany, NY commuters found that those whose work trip was made at lower than average speed were more likely to carpool, at least for trips of more than 10 miles (Brunso *et al.*, 1979). The effects of other transportation characteristics, such as the availability and quality of public transportation, have not been systematically examined.

How commuters carpool has been less thoroughly analyzed, although the results of the studies are more consistent. A significant portion of carpools are known to be composed of household members. A U.S. study found that 35% of all carpools were composed of household members only (Kendall, 1975), an Australian study (in Melbourne) found that about three out of eight carpools had this composition (Richardson and Young, 1982), and Hartgen (1977) has estimated that one-third of all carpoolers commute with household members, although he cites a range of values from various studies. Studies have also found significant differences in trip distance and trip circuity between household carpools and those composed of unrelated individuals (Kendall, 1975; Richardson and Young, 1982).

Two studies indicate that a significant amount of carpooling is actually "car-sharing," that is, the sharing of a single vehicle by two or more commuters and not a ridesharing arrangement in which each commuter regularly contributes his/her vehicle to the pool. A U.S. study determined that two-thirds of all unrelated carpoolers either drive or ride all the time (Kendall, 1975). In Great Britain, about one-third of carpool applicants are offering rides, another one-third are seeking rides and only one-third wish to pool their vehicle and share driving responsibilities (Bonsall, 1982).

Research has also shown that carpools are flexible commuting arrangements which do not operate every day, either at full strength or even at all (Attanucci, 1974). In addition, most carpools composed of unrelated individuals are destination-oriented carpools; that is, the members work at the same location (Richardson and Young, 1982). Neighborhood-oriented carpools, composed of friends or neighbors who live in proximity to one another but have somewhat dissimilar destinations, are much less common.

Why workers carpool is the subject of considerable disagreement. The earliest major study of carpooling identified economic factors as the primary cause of carpooling, but also paved the way for different interpretations by identifying strong psychological barriers to ridesharing (Voorhees, 1973). Several subsequent studies have concluded that carpooling propensity is affected more by attitudinal factors than socio-demographic attributes or the standard travel time and cost variables used in transportation mode choice models (Hartgen, 1977; Margolin and Misch, 1978; Duecker *et al.*, 1977; Horowitz and Sheth, 1978). Much attention has been devoted to measuring how commuters perceive carpooling and driving alone in terms of underlying convenience factors, with the general finding that commuters are reluctant to carpool because it is inconvenient on several dimensions and limits personal independence. The one study which examined the interactive effects of trip characteristics, socio-demographic factors, and attitudes towards convenience attributes concluded, however, that carpooling propensity is affected by household vehicle availability (as measured by the ratio of licensed drivers to vehicles) as well as attitudes (Brunso *et al.*, 1979).

There are two major exceptions to the attitudinal

Table 1. Key variables potentially affecting carpooling propensity

Sociodemographic	Transportation	Locational
Age	Travel Distance/Time	Urban Population
Sex	Travel Speed	Residential Location (metropolitan vs. nonmetropolitan, suburb vs. central city)
Income	Trip Cost	
No. of Household Workers	Auto Availability	Employment Location (suburb, central city, CBD)
	Transit Availability/ Quality	

explanation of carpooling propensity. A study for the Federal Energy Administration adopted the approach that carpooling could be predicted almost entirely by the use of transportation, locational and socio-demographic variables through a utility maximization framework using a logit model (Cambridge Systematics, 1976). In addition, the authors hypothesized that carpooling's attractiveness increases with the number of workers in the household (due to the opportunity for household carpooling in multiple worker households), that its attractiveness decreases as the ratio of vehicles to licensed drivers increases and that the primary worker in a household is less likely to carpool that other household members when there are fewer vehicles than workers. In fact, the model developed *was* able to accurately predict the carpooling mode share in Washington, D.C. and contained variables with statistically significant coefficients whose signs and values were consistent with the above hypotheses.

A Canadian study also eschewed the attitudinal approach to explaining carpooling, focusing on the socio-demographic and tripmaking characteristics of auto passengers (McCoomb and Stewart, 1982). This study found that auto passengers were more similar to transit users than to auto drivers, and that sociodemographic factors could be used to distinguish among the three groups, albeit with only moderate success.

Research Implications of Previous Studies

The studies reviewed above do not cumulate to either a definitive or a comprehensive picture of carpooling. In particular, the studies have not reached common conclusions about which socio-demographic, transportation, and locational variables have the greatest influence on carpooling propensity, nor have they offered a consistent description of the type of individuals most likely to carpool. A notable deficiency is the limited analysis of the economic factors which influence carpooling, although the demonstrated effects of trip distance and vehicle availability on mode choice indicate that economic motivations may be critically important in understanding carpooling.

In addition, most of the prior studies offer only partially reliable insights into carpooling due to problems of representativeness of the data used. With a single exception, the data for the American studies have come from one or a few urban areas (three is the maximum) and in most cases do not even represent a random sample of the urban area in question, being skewed due to study objectives or problems of data availability. The exception is a 2000 person national sample collected by the National Opinion Research Center which included approximately 500 carpoolers (Kendall, 1975). However, this study included only automobile commuters, not transit users as well, and did not include any data on actual trips. Consequently, these studies are best considered to be interesting exploratory examinations of carpooling behavior, suggesting hypotheses and avenues for inquiry which can be best analyzed with a more satisfactory data set.

This reveiw of the previous research on carpooling suggests guidelines for conducting the present analysis. First, the data sample must be national in scope, include users of all modes, and contain information on actual travel, as well as complete transportation and socio-demographic data. Second, in determining who carpools and why, the analysis should focus on the sociodemographic, transportation, and locational variables listed in Table 1, to the extent that such information is available in the data set. Third, in analyzing how commuters carpool, attention should be focused on the composition of carpools (household members vs. unrelated persons), the size of the carpool, the regularity of the carpool arrangement and how carpool members share (or do not) vehicles.

DATA AND METHODOLOGY

An ideal data set was available to overcome the data limitations of previous carpooling studies, namely the 1977–78 Nationwide Personal Transportation Survey (NPTS). The NPTS sample contains more than 22,000 workers, including more than 3000 individual carpoolers. This is by far the largest number of carpoolers included in any random survey of commuters to date. The NPTS consists of a representative national sample of over 18,000 households. Personal interviews with members of these households were conducted by the Census Bureau between April, 1977 and March, 1978. All household members were asked to report detailed travel information for each trip they made within a 24-hr period. The NPTS questionnaire also gathered demographic, employment and journey to work data for

each household member. For each household, information was obtained on vehicle ownership and vehicle characteristics, as well as public transportation availability and proximity. No attitudinal data as such is included in the survey, although commuters who drove alone to work were asked why they did not carpool.

The entire NPTS sample was used to construct contingency tables and compute mean values for variables. In addition, approximately equal samples of drive alone, carpool, and transit commuters (850 to 900 workers each) were selected for more detailed statistical analyses.

One possible objection to the use of the NPTS as a data set is that 1977–78 data is not representative of current urban transportation conditions affecting carpooling behavior. In particular, the nominal price of gasoline in 1985 is about twice as much as in 1978, and traffic congestion has increased in many large metropolitan areas during the past several years. However, the 1985 real price of gasoline is only about 15% greater than in 1977–78, and substantial improvements in automotive fuel economy during this period have negated even this difference. As for congestion effects, the NPTS data indicate that automobile commuting speeds have at most a weak association with carpooling propensity.

The comprehensiveness of the NPTS data allows it to be used to examine a number of assertions and hypotheses previously developed about carpooling, as well as to obtain definitive information on who carpools and how they do so. Three major hypotheses are examined in this paper.

(1) Carpooling is *not* a homogeneous commuting mode, but rather a set of submodes whose users have distinctly different characteristics.

(2) Carpooling propensities can be predicted, to at least a certain extent, through the use of particular transportation and socio-demographic variables. Nonattitudinal variables other than travel distance are hypothesized to have a significant influence on carpooling, and economic variables in particular are hypothesized to be important.

(3) A major increase (e.g. 25% or more) in overall ridesharing is unlikely given the transportation and situational characteristics of carpoolers and drive alone commuters.

Several elementary statistical techniques—descriptive statistics, contingency analysis, *t*-tests, and correlation analysis—represent the principle method of conducting this analysis. In addition, discriminant analysis is used in an attempt to determine if certain variables can satisfactorily distinguish carpoolers from commuters who drive alone or use transit.

WHO CARPOOLS? THE CHARACTERISTICS OF CARPOOLERS

Definitional Issues

Before examining the characteristics of carpoolers, it is necessary to determine just who constitutes a carpooler. Some analysts adopt a restrictive definition, considering carpoolers to be only unrelated individuals who share responsibility for vehicle provision and driving. Other analysts distinguish between "external" carpools (unrelated individuals) and "internal" (household) carpools (Richardson and Young, 1982). Yet other analysts distinguish between carpooling and car-sharing, but consider household car-sharing to be a legitimate form of ridesharing (Bonsall, 1982). The definition used here is purposefully broad—a carpooler is considered to be anyone who shares transportation to work in a private vehicle with another worker.

Defined in this way, 17.1% of the commuting workers (excluding those who work at home) in the NPTS sample are carpoolers, and 18.5% of the workers who commute in a vehicle (excluding bicyclists) are carpoolers. Carpoolers represent 19.6% of all *private* vehicle commuters (excluding bicyclists). For the remainder of this paper, all carpooling percentages will be expressed in terms of vehicular commuters excluding bicyclists, unless otherwise stated.

Carpooling does not represent a singular mode of commuting, but rather encompasses distinctly different commuting arrangements. Three distinct types of carpoolers were delineated, based on previously developed categorizations as well as consideration of the important difference between pool arrangements in which each commuter contributes a vehicle and driving assistance and those in which they do not. (1) *Household carpoolers*, who commute together with at least one other worker from the same household. (2) *External carpoolers*, who share trans-

Table 2. Disaggregation of carpoolers

Type of Carpooler	Portion of All Carpoolers	Portion of all Vehicular Commuters
Household Carpool (HHCP)	42.3%	7.8%
External Carpool, Drive Only or Share Driving (EXTCP)	36.1	6.7
External Carpool Riders (CPRDR)	21.6	4.0
All types	100%	18.5%

portation with unrelated individuals and who either share driving responsibilities or drive always. (3) *Carpool riders*, who commute with other unrelated workers but who ride only and never provide a vehicle. These three different groups of carpoolers will be referred to as HHCP, EXTCP and CPRDR, respectively. Table 2 gives the relative size of these groups.

Salient Characteristics of Carpoolers and Users of Other Modes

Table 3 presents salient socio-demographic and transportation characteristics of drive alone, carpool and transit users, as well as illustrating how modal use is affected by these six factors. As can be seen, the carpoolers differ significantly from the users of the other two modes on every variable. Compared to drive alone commuters, they make longer trips, have lower incomes, are more likely to be married females come from households with more workers and which own fewer vehicles per worker and experience a greater commuting cost to income (CCTI) burden if they were to drive alone. The commuting cost to income (CCTI) variable was created by assuming that out-of-pocket commuting costs consist of gasoline costs plus parking charges, and is computed as if each commuter drove alone. Gasoline costs were estimated at 4¢ per mile, based on a gasoline cost of 60¢ per gallon and a vehicle fuel efficiency of 15 miles per gallon, representative values for 1977–78. Parking costs were actual daily parking charges as reported by NPTS respondents. Household income was taken as the midpoint of the categorized NPTS income range, and was divided by the number of workers in the household to create an income per worker variable. A more satisfactory measure of commuting cost burden would use actual gasoline costs, not averages, as long distance commuters are more likely to drive vehicles with above average fuel efficiency. However, the NPTS did not include comprehensive data on vehicle fuel efficiency, so average values had to be used.

Striking differences in transportation and demographic characteristics among users of the carpool submodes are equally evident. The CPRDR group has significantly lower income and fewer vehicles per worker than the other two carpool sub-groups, while the EXTCP group has much longer work trips and much higher levels of automobile availability than the other two groups. In addition, females are a much lower portion of the EXTCP group than the others. The major distinguishing characteristics of

Table 3. Mode and submode use vs. transportation and household characteristics

	HHCP	EXTCP	CPRDR	(Total Carpool)	Drive Alone	Public Transit	Total
Overall Mode Share	7.8%	6.7%	4.0%	(18.5%)	75.1%	6.5%	100.0%
Trip Distance							
≤ 5 mi.	7.2%	3.7%	4.3%	(14.2%)	78.3%	6.5%	100.0%
≥ 15 mi.	8.0%	12.2%	4.0%	(24.2%)	69.7%	6.1%	100.0%
≥ 25 mi.	9.0%	20.4%	4.9%	(34.4%)	57.5%	7.8%	100.0%
Average	10.4mi.***	17.4 mi.***	9.8 mi.***	(12.8 mi)*	9.0 mi.	11.2 mi.	
Household Income							
≤ $10,000	8.5%	6.2%	7.2%	(21.9%)	67.9%	10.1%	100.0%
≥ $20,000	7.7%	6.2%	2.0%	(15.9%)	78.6%	5.5%	100.0%
Average	$18,321***	$18,020***	$13,628**	($17,198)*	$18,932	$16,210	
Drive Alone Out of Pocket Commuting Cost/Income per Worker (CCTI)							
≤ 0.5%	3.8%	3.3%	3.6%	(10.7%)	84.6%	4.7%	100.0%
≥ 3.0%	5.2%	5.0%	3.4%	(13.6%)	80.0%	6.4%	100.0%
≥ 5.0%	10.0%	14.7%	6.9%	(31.6%)	61.7%	7.7%	100.0%
Average	.036**	.050***	.047***	(.044)*	.025	NA	
Household Vehicles per Worker							
< 1.0	15.8%	4.6%	10.3%	(30.7%)	50.5%	18.8%	100.0%
1.0	6.4%	7.4%	2.5%	(16.3%)	80.1%	3.7%	100.0%
> 1.0	3.7%	6.8%	1.4%	(11.9%)	86.5%	1.6%	100.0%
Average	.88**	1.21**	.67**	(.95)*	1.26	.55	
Workers in Household							
1	0	6.9%	4.4%	(11.3%)	79.4%	9.3%	100.0%
2+	12.0%	6.5%	3.65	(22.1%)	72.9%	5.1%	100.0%
Average	2.29**	1.54***	1.53***	(1.71)*	1.52	1.39	
Sex and Marital Status							
Unmarried Male[a]	4.6%	5.3%	3.7%	(13.6%)	76.2%	10.2%	100.0%
Unmarried Female[a]	4.1%	5.3%	7.8%	(16.3%)	68.1%	15.5%	100.0%
Married Male	6.5%	7.8%	2.4%	(16.7%)	78.9%	4.4%	100.0%
Married Female	11.3%	5.5%	4.6%	(21.2%)	73.9%	4.8%	100.0%

[a]Head of Household only.

*Significantly different at .01 level from drive alone and public transit mean values.
**Significantly different at .01 level from mean values for both other carpool submodes.
***Significantly different at .01 level from mean value for one other carpool submode.

Table 4. Carpooling and size of metropolitan area

Modal Share	Population of Metropolitan Area: Non-SMSA	less than 250,000	250,000-500,000	500,000-1,000,000	1 million or more
Carpoolers as Percent of all Commuters	20.6%	18.8%	18.4%	19.1%	16.2%
Carpoolers as Percent of Auto Commuters	20.8%	19.2%	19.1%	20.0%	18.6%
EXTCP as Percent of Auto Commuters	7.7%	6.7%	6.8%	6.4%	6.0%

the HHCP group are a high number of workers in the household and a significantly lower commuting cost burden compared to the other carpool sub-groups.

The users of carpool submodes can also be compared to the drive alone and transit commuters. EXTCP users differ significantly from drive alone commuters only on the trip distance and commuting cost burden variables, whereas the other two types of carpoolers differ from drive alone commuters on almost every *other* variable, but are similar in travel distance. In addition, the carpool riders bear some important similarities to transit users, notably with respect to income, vehicles per worker, trip distance and the overrepresentation of unmarried females.

Each of the factors included in Table 3 has a pronounced impact on carpooling propensity for at least one of the carpool submodes. The three major influences on *overall* carpool mode share are trip length, auto availability and the commuting cost burden (which is significantly correlated with trip length). Somewhat surprisingly, parking cost does not appear to be a major influence on overall carpool mode share. However, such a small percentage of commuters in the NPTS survey paid to park (3.5% of all automobile commuters and less than 4% of all carpoolers) that these results should be interpreted cautiously.

Relative to carpool submodes, long trip distances and a high commuting cost burden result in striking increases in EXTCP carpooling, and the presence of parking charges also demonstrates some relationship with this type of carpooling. In contrast, HHCP and CPRDR usage is primarily associated with fewer household vehicles than workers. CPRDR users are also more likely to be drawn from the ranks of lower income workers and unmarried female heads of households. Married females, on the other hand, are disproportionately represented among household carpoolers.

Carpooling and the Size of Urban Areas

An assertion that has been made by prior studies is that the carpool mode share is highest in large metropolitan areas (Oppenheim, 1979; Hartgen, 1977). The NPTS data refute this hypothesis. As Table 4 indicates, the percentage of carpoolers is remarkably stable across different sizes of urban areas. This result holds no matter how carpooling is defined, although a small reduction in total carpooling mode share for metropolitan areas of one million or more population is indicated. The overall finding is that population size per se has little effect on carpooling propensity among automobile commuters.

HOW DO COMMUTERS CARPOOL?

Carpool Composition, Size, and Roles

The NPTS data confirm that a substantial portion of carpoolers are members of household carpools, over 40% of the carpoolers in the sample. Most of these household carpools are composed only of household members, although about 10% contain at least one unrelated individual. Overall, these "mixed" carpools comprise about 5.5% of all carpoolers.

Although carpools comprised only of unrelated persons comprise 58% of all *carpoolers*, they represent only 54% of all *carpools*. This occurs because these carpools tend to be larger than household carpools. Purely household carpools are overwhelmingly composed of two persons; 95% have only two members. Mixed carpools by definition have at least three members. Internal carpools have an overall average of 2.25 members. In contrast, external carpools average 2.63 members, and 38.5% of all such carpools have three or more members.

The NPTS results also confirm that the majority of carpooling does not involve shared driving arrangements. Among EXTCP carpoolers (the only type other than "mixed carpools" where shared driving is either meaningful or feasible), 40% share the driving responsibilities (and presumably contribute their vehicle when doing so), 39% ride only and 21% drive only. Among external carpoolers from households with at least as many vehicles as workers, however, there is a marked increase in shared driving arrangements, to 51%. Only 23% of carpoolers in this latter group are riders all the time.

Regularity of Carpooling

As might be expected for a flexible commuting arrangement such as carpooling, carpools are not

Table 5. Regularity of carpool arrangement

Type of Carpool	% of carpoolers not carpooling on survey day	% of carpools less than nominal size on survey day
External	14.9%	20.1%
HH members only	42.7%	43.8%
Mixed	14.1%	40.0%
All types	22.2%	27.0%

always in operation. The NPTS questionnaire not only asked workers how many persons were members of their carpool, but also contains a one day trip diary which, among other items, records information on the number of persons in the vehicle on every automobile trip. It was thus possible to compare typical commuting behavior with the behavior which actually occurred on a particular day. This comparison yielded the needed information on the regularity of carpooling.

According to the NPTS data, on a typical day more than one out of five nominal carpoolers do not carpool (See Table 5). Among household carpools the percentage is strikingly higher, with 43% of such carpoolers not carpooling on the survey day.

Because information is available only on carpoolers, not carpools, it is not possible to determine how many carpools are not in operation at all on a typical day. It is possible, however, to determine the percentage of carpools which on the survey day had less than the nominal number of members, and this is shown in Table 4. Only 20% of all external carpools operated with a missing member on the survey day, compared to 40% or more for the other two types of carpools.

These results indicate that external carpools operate in their nominal mode about four out of five days per week, and can be considered to have a high degree of regularity. They tend to correspond to the typical view of a carpool as a relatively fixed commuting arrangement, albeit one with sufficient flexibility to permit members to opt out for a day when required by individual needs.

On the other hand, internal carpools represent a much more flexible commuting arrangement. This appears to be a consequence of two factors. First, many married women who are household carpoolers are apparently either part-time workers or have family responsibilities involving chauffering children which prevent them from taking part in the household carpool on a regular basis. This can be inferred from the fact that 65% of this group did not take a work trip on the survey day, compared to 30 to 40% of household carpoolers from other household status categories. (Because the survey was conducted on all seven days of the week, a missing rate of 25 to 30% is to be expected given the typical Monday to Friday work week.) But when one member of a household carpool drops out for a day, the carpool almost always dissolves, given that very few have more than two members. Thus it is not surprising that married men who are nominally members of household carpools actually engage in carpooling on only 50% of their work trips.

A second plausible reason that internal carpooling exhibits less regularity than external carpooling is that it is less motivated by economic factors. The mean CCTI value for internal carpoolers is over 25% less than for external carpoolers. Particularly for households with at least one vehicle per worker, internal carpooling is probably motivated as much by convenience factors as economics. Such carpoolers should logically be more likly to dissolve the carpool for the day (particularly as their decision will be easily understood by the other carpool member) when it is inconvenient to maintain the arrangement.

WHY DO COMMUTERS CARPOOL?

Commuting Cost Burden

As noted previously, carpooling propensity increases sharply with increases in both trip distance and the CCTI variable (which is strongly correlated with trip distance). The carpool mode share increases from 15.5% for trips of 10 or less miles to 33% for trips of more than 25 miles, while it increases from 12% for a CCTI ratio of 1.5% or less to 26.5% for CCTI ratio of more than 3%. In addition, carpool riding is most prevalent among workers in the lower quartile of the income distribution. These findings indicate rather convincingly that those workers who stand to reduce their commuting cost burden the most by carpooling are in fact those most likely to do so.

This commuting cost burden explanation applies to all three modes of carpooling, but is particularly applicable to the EXTCP submode. These carpoolers represent 16% of all automobile commuters whose CCTI ratio exceeds 5%, compared to only 5% of automobile commuters whose CCTI ratio is 3% or less.

Vehicle Shortages

The NPTS data also provide convincing evidence that carpooling propensity is strongly influenced by vehicle availability. The vehicles per worker (VPW) variable demonstrates the strongest correlation with carpool mode share of all the factors considered in this analysis. Among households with fewer vehicles than workers, 38% of all automobile commuters are carpoolers, compared to only 15% when the VPW ratio is at least 1.0. Nearly 60% of carpool riders

come from households with fewer vehicles than workers, as do 46% of all household carpoolers.

Because vehicle shortages are not shared equally among household members, there is a pronounced pattern to carpooling in households which have multiple workers and fewer vehicles than workers. In this situation, carpool riders are drawn disproportionately from the ranks of married women and other nonheads of household. In such households, 9.6% of married women commute by carpool riding (10.2% of other nonhousehold heads) compared to only 3.8% of married men. When multiple worker households have at least as many vehicles as workers, however, the propensity for carpool riding drops sharply, as does the married male to married female differential—1.3% of the former and 2.2% of the latter commute by carpool riding (3.0% of other nonhousehold heads).

These results tend to confirm the hypothesis of unequal vehicle sharing in multiple worker households with vehicle shortages. The male head of household typically has first claim on a vehicle; this leaves other household members to work out their own commuting scheme when a household carpool is not feasible. For more than 35% of non-household heads in this situation, the solution is riding in a carpool with unrelated individuals; transit is the solution 49% of the time. In contrast, 88% of all married men from such households commute in a household vehicle.

Economic Factors and Carpooling

The above results suggest that economics plays a major role in carpooling. The commuters most prone to carpool are those making long trips, those with relatively high commuting cost burdens if they drive alone, and those from households with fewer vehicles than workers, a condition which can only be alleviated by higher household transportation expenditures. The importance of economic factors can be more graphically illustrated by considering these variables jointly. All commuters can be classified in one of four categories based on their values for VPW and CCTI. The classification was made on the basis of whether VPW was less than 1.0 or greater than or equal to 1.0, and whether CCTI was greater or less than 2.5%. The effect of this classification scheme on mode choice is shown in Table 6, *where the carpool percentage is that for automobile commuters only*.

Table 6 clearly indicates the important role economic motives play in carpooling, and the relative importance of the two factors used here. When *both* economic motivations are present, carpooling is *four* times more likely than when neither is present (41.5% vs. 10.7%). When only one motivating factor is present, carpooling is still two to three times more likely than when neither of those economic motivations exists.

Table 6 also demonstrates the relative influence of commuting cost burden and vehicle availability on the three sub-modes of carpooling. Household carpooling and carpool riding are primarily a response to vehicle shortages, whereas external carpoolers are motivated primarily by the commuting cost burden. This makes much sense, as purchasing an additional vehicle for commuting purposes will usually be much more expensive on an annual basis than the cost difference between driving alone and carpooling. Since 84% of all external carpoolers come from households with VPW $\geq$ 1.0, compared to only 54% of household carpoolers and 41% of carpool riders, the large majority of EXTCP users have already made the monetary outlays necessary to bring household vehicles into balance with household workers. In contrast, many HHCP and CPRDR users must still overcome this economic hurdle. Nearly 50% of the latter two groups are making trips of five or less miles, so their out of pocket savings from carpooling are small indeed compared to the vehicle ownership cost savings associated with carpooling.

Finally, Table 6 indicates the major problem confronting attempts to increase ridesharing. Five out of nine of all commuters have relatively inexpensive work trips and do not experience household vehicle shortages, a combination of circumstances which results in a level of carpooling 45% below the overall national average. Only one out of twelve commuters experience the combination of economic motivations which lead to a very high level of carpooling.

Multiple Worker Households and the Convenience of Carpooling

Carpooling is about 50% more prevalent among multiple worker households than those with only one worker. Obviously, the presence of multiple workers within a household makes carpool formation much less problematic than is the case otherwise, *provided* that the employment destination of the household

Table 6. Economic motivations and carpooling propensity

	Out of Pocket Commuting Cost/ Income per Worker			
	less than 2.5 percent		greater than 2.5 percent	
Mode	vpw < 1.0	vpw ≥ 1.0	vpw < 1.0	vpw ≥ 1.0
Drive Alone	68.2%	89.3%	58.5%	78.1%
HHCP	13.1%	3.8%	18.2%	5.8%
EXTCP	4.0%	5.3%	9.4%	13.0%
CPRDR	14.6%	1.6%	13.8%	3.1%
(Total Carpool)	(31.8%)	(10.7%)	(41.5%)	(21.9%)
% of all commuters	12.9%	55.5%	8.7%	22.9%
% of all carpoolers	18.6%	34.3%	18.2%	28.8%

members are in reasonable proximity. Richardson and Young (1982), in a study of carpooling in Melbourne, Australia, found that the average work end trip deviation for internal carpools was 1.1 miles, although 20% had to deviate at least 2.5 miles. If these results are indicative of the U.S. situation, they imply that most household carpools do not have to endure inconvenient deviations.

In addition, carpooling is evidently much more convenient than transit for workers from households which own at least one vehicle, but have fewer vehicles than commuters. For workers from such households who do not drive alone, transit represents 22% of all commuter trips, compared to 49% for household carpooling and 78% for all forms of carpooling. In contrast, among workers from households with *no* vehicles, 70% of all vehicular commuters use transit. Whatever the inconvenience entailed in making destination-end deviations for a household carpool, for most commuters this inconvenience is apparently of a lesser magnitude than that associated with transit use.

Transit and Carpooling

The dominance of the drive alone mode for work trips, particularly among commuters from households with at least as many vehicles as workers (83% drive alone among this group), indicates that both carpooling and transit are considered to be inferior commuting choices by most workers. Which of these "second best" modes is actually used by commuters who do not drive alone is, in the aggregate, strongly influenced by the quality and availability of transit.

Transit is most likely to be unavailable or of poor quality in non-metropolitan areas and small metropolitan areas (those with a population of less than 250,000). Among workers from households with VPW < 1 who live in such areas, only 11% said transit was available for their work trip. Among workers from these vehicle short households who did not drive alone, 91% commuted in a carpool and 40% commuted as a carpool rider—the obvious alternative to transit for workers from such households when a household carpool is not feasible. In contrast, among households with fewer vehicles than workers residing in SMSAs of more than 1 million population, 66% said transit was available for their work trip, only 38% who did not drive alone commuted in a carpool, and only 15% were carpool riders. Transit is the mode of choice for 70% of vehicle short commuters in large metropolitan areas. It is apparent, therefore, that carpooling picks up the slack for transit in environments where the latter is of poor quality.

The effects of *high* quality transit service on carpooling are not easily analyzed by the NPTS data, as it is not possible to identify the specific urban area in which survey respondents resided and then relate measures of transit availability to carpooling and transit use. However, 1980 Census data can be used to illustrate how carpooling is affected by the aggregate quality of transit in large (more than 1 million population) metropolitan areas.

Table 7 reveals that in the most transit-oriented large cities (defined as those with a transit mode share for work trips of at least 12%), the carpool mode share is only 90% of that in all other large SMSAs. The portion of *automobile commuters* who carpool in the high transit use cities, however, is greater than in the other large SMSAs.

These results suggest a multi-faceted relationship between carpooling and transit use. Transit is a more appealing alternative to driving alone than carpooling *if* reasonably high quality transit is available, presumably reflecting transit's substantial temporal flexibility advantage over carpooling in such environments. Moreover, the higher precentage of carpooling among automobile commuters in these SMSAs is likely to result from the same environmental conditions which are conductive to high transit use—relatively dense population and concentrated employment patterns in central areas. These conditions create both high employment densities and high parking charges in central areas, two factors promoting carpooling. In fact, the Census data indicate that the higher carpool rate among automobile commuters in high transit use cities stems largely from a high rate for trips to central areas. The carpooling rate among automobile commuters for CBD trips is 28% greater in the high transit use SMSAs than in the other large SMSAs, whereas it is virtually the same for suburban work trips in the two sets of cities.

Of course, transit quality in *any* SMSA would be

Table 7. Modal shares in different types of large SMSAs

	Mode Shares				
Type of SMSA	Drive Alone	Carpool	Public Transit	Carpool/ Total Auto	Carpool/Non-Drive Alone
Low-Medium Transit Use (n=29)	73.4%	20.2%	6.4%	21.6%	75.9%
High Transit Use (n=9)	56.1%	18.1%	25.8%	24.4%	41.2%
Difference Between High and Low Transit Use SMSAs	-23.6%	-10.4%	+303.1%	+13.0%	-45.7%

expected to be best and the conditions promoting transit and carpool use the most pronounced for work trips to CBD destinations. In SMSAs with a transit mode share of less than 4%, carpooling increases substantially among both all workers (25% increase) and automobile commuters (45% increase) for CBD trips compared to SMSA level mode choices, even as the transit mode share increases as well. Although carpooling among automobile commuters for CBD trips increases by an even greater amount (68% increase over SMSA level) in the high transit use cities (presumably reflecting stronger motivating conditions), it appears that the intra-regional environmental influences on carpooling propensity are present, albeit of different strength, in nearly all large urban areas.

The relationship between transit and carpooling use is a complex one, affected by many specific local conditions, and this analysis barely scratches the surface. The data do indicate, however, that depending on the nature of the urban environment, transit and carpooling can be complementary or substitutable modes.

DISCRIMINATING BETWEEN CARPOOLERS AND OTHER COMMUTERS

The previous analysis has demonstrated that several factors have an important influence on the propensity of carpooling. Can these factors be used to predict which types of commuters will carpool? As a means of answering this question, a series of discriminant analyses were performed on the NPTS data. Discriminant analysis was used rather than a logit type mode choice model because insufficient information was available on the characteristics of the commuting choices facing the workers in the NPTS sample (in particular, travel times and costs by alternatives to the chosen mode were not known). Using a subset of the entire data set, consisting of 850 to 900 commuters from each of the drive alone, carpool and transit modes, attempts were made to discriminate between users of these modes (as well as the carpool sub-modes) on the basis of linear discriminant functions containing several or all of the variables in Table 8. (A transit availability variable could not be used because it was not a continuous variable, which is a requirement for discriminant analysis.)

Table 9 presents a summary of the results. Most of the canonical correlations clustered between 0.50 and 0.55, and most of the Wilks lambda's clustered around 0.70. These statistics indicate relatively weak discriminating power for the variables used. The classification results also left much to be desired, although the discriminant functions performed considerably better than chance in correctly classifying modal users—overall error reduction ranged from 35 to 50%, albeit less for the carpool modes. On the other hand, the univariate F ratio was robust for all of the variables included in the discriminant functions, indicating that these variables do satisfactorily separate carpoolers from other commuters. However, the lackluster showing on the other statistics indicates that the modal groups do not cluster tightly around the mean values for the discriminating variables.

The discriminant analysis indicates that the most important factors influencing carpool mode choice for the HHCP and CPRDR submodes are vehicles per worker, the number of workers in the household and either trip distance or CCTI, depending upon the selection criteria used. The standardized discriminant coefficient for VPW was at least three times as large as for any other variable. For the EXTCP submode, the most important dicriminating variable is trip time or distance, followed by CCTI.

The problem confronting the discriminant analysis, as it would any technique which attempts to predict carpooling propensity from these variables, is that even in the most carpool-prone categories the majority of workers drive alone. For example, even though a very high *relative* percentage of carpooling occurs among workers with a CCTI ratio of more than 5%, nearly 60% of workers in this category drive alone. Consequently, this variable cannot accurately discriminate between the users of two modes. Adding other variables is only moderately helpful, as no combination of criteria which yield at least 5% of the workers in the sample can also obtain a carpool mode share of at least 50%. Discriminant analysis can certainly indicate which variables are important in influencing carpooling and specify their relative importance, but it does not yield highly accurate predictions of which workers will carpool.

These results offer partial support for those researchers who have emphasized attitudes as a major determinant of carpooling. Although it is evident from this analysis that carpooling tendencies are created by an individual's transportation situation and socio-demographic status, these factors alone are insufficient to explain why workers carpool. For ex-

Table 8. Variables used in discriminant functions

1.	Vehicles per worker	(VPW)
2.	Workers in the household	(WKERS)
3.	Drive alone out of pocket commuting cost/income per worker	(CCTI)
4.	Trip distance	(WKDIST)
5.	Travel Time	(WKTIME)
6.	Family Income	(FAMINC)
7.	Income per worker	(INCPWKR)

Table 9. Results of discriminant analyses

Modes	Discriminating Variables and Relative Importance	Canonical Correlation*	Wilks Lambda*	% Correctly Predicted All	% Correctly Predicted Carpool
DA,CP,PT	VPW, WKERS, WKDIST[a] INCPWKR, CCTI[a]	.50	.73	52.4% (33.3)[b]	44.2% (33.3)
DA,HHCP, EXTCP, CPRDR,PT	Same as Above	.51	.68	40.9% (20.0)	31.3% (20.0)
DA,HHCP, PT	Same as Above	.54	.67	56.3% (33.3)	45.5% (33.3)
DA,CPRDR, PT	Same as Above	.54	.70	60.3% (33.3)	39.7% (33.3)
DA,HHCP	VPW, WKERS, CCTI	.32	.90	63.6% (50.0)	71.0% (50.0)
DA,EXTCP	WKTIME, FAMINC, CCTI	.23	.94	69.1% (50.0)	51.0% (50.0)

*Value for best discriminant function

[a]Importance of these variables could be reversed depending upon selection criteria used.

[b]Percent expected by chance

ample, *one-third* of all carpoolers do *not* face a lengthy commute, a significant commuting cost burden, or a vehicle availability problem. Given the association which has previously been demonstrated between carpooling and attitudes towards this mode, it logically follows that attitudes are another important factor in determining which commuters carpool.

CONCLUSIONS

The NPTS data set permits the first representative and comprehensive analysis of the socio-demographic, transportation and situational factors which influence carpooling. The analysis of the NPTS data largely confirms the three hypotheses stated at the outset. First, there *are* major differences between the users of the three carpool submodes considered here. Significant differences in transportation situation, economic status, household size, vehicle availability and regularity of carpooling distinguish these user groups from each other, and refute any notion that carpooling can adequately be considered as a homogenous commuting mode.

Second, the analysis reveals, contrary to the results of some previous studies, that nonattitudinal factors are indeed important in determining who carpools—there is a major element of rational choice in the mode selection process. Those factors which one would expect to influence carpooling propensity—trip length, commuting cost burden, automobile availability, the number and status of workers in the household, trip destination and the quality and availability of public transportation—have been shown to do so. Nonetheless, these factors are not so powerful that they enable us to predict with high accuracy who will carpool, except possibly when incorporated into a full-blown demand model of mode choice (see Cambridge Systematics, 1976, for an example). There thus remains an important role for individual attitudes towards carpooling and driving alone to influence the choice between these modes, although it seems likely that such attitudes operate *within* the more objective influences which have been indentified.

Finally, this analysis indicates the extreme difficulty of stimulating a major increase (e.g. 25% or more) in the amount of carpooling which occurs in an urban area. A relatively small percentage of drive alone commuters are likely to be realistic candidates for carpooling. Only 15% have fewer vehicles than workers in their household, only 15% are making trips of more than 15 miles, and only 20% have a CCTI ratio of more than 3%. Although 34% possess one of these three characteristics, this means that fully two-thirds of all drive alone commuters lack any major transportation or economic motivation to carpool. Moreover, even among those who might be motivated to carpool, the majority cite either irregular work hours or location (44%), or the need for their vehicle or a desire for independence (15%) as the reason for not carpooling. These obstacles to carpooling are very difficult to overcome.

Organized ridesharing programs are aimed, of course, at both promoting carpooling and removing obstacles associated with the difficulty of finding carpool partners (34% of the drive alone commuters in the NPTS cited the lack of a potential carpool partner as the reason for not pooling). Such programs, however, are oriented towards the type of commuter which is best represented by EXTCP carpoolers, but such workers comprise less than 40% of all current carpoolers. Moreover, there is very little that or-

ganized ridesharing programs can do which will increase HHCP and CPRDR carpooling. The former depends upon households having multiple workers with reasonably proximate employment destinations, and the latter occurs primarily in response to vehicle shortages, which are in long term decline.

To achieve a 25% increase in carpooling—which represents a 60% increase in EXTCP carpooling if all newly formed carpools are of this type—over one-half of the drive alone commuters with the most time consuming (distance greater than 25 miles) and economically burdensome (CCTI greater than 5%) work trips would have to be converted, or more than one-sixth of the drive alone commuters with *any* motivating characteristics would have to be converted. While such outcomes are not impossible, they are certainly far beyond the achievements of any areawide organized ridesharing campaigns to date.

The problem in substantially increasing carpool mode share is much more fundamental than unfavorable attitudes towards carpooling. The large majority of commuters are simply unmotivated to carpool due to relatively short trips, relatively low commuting cost burden and relatively high levels of vehicle availability. When important motivating factors—vehicle shortages, long trips, high commuting cost burden—are present, a significant portion (at least 20 to 30%) of all vehicular commuters engage in carpooling without any organized assistance. This fact in no way negates the need for either employer-based or areawide carpooling programs, but it does suggest that they are not likely to lead to major increases over the "natural" level of carpooling, that is, the level which occurs without widespread formal organization. This may be a sobering conclusion to the most optimistic advocates of ridesharing, but the past decade of carpooling activity has not produced evidence to indicate that is it too pessimistic a conclusion.

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